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A transparent financial risk factor library

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Abstract

Financial risk factor time series are essential for empirical asset pricing research, yet existing factor libraries offer limited reproducibility, lack modularity, and target single programming languages. We present a fully replicable, multilingual financial risk factor library that addresses these shortcomings. Our modular framework covers the full pipeline from raw data to factor returns: data acquisition, cleaning, variable construction, and portfolio aggregation. The library provides both an application programming interface (API) and native implementations in Python and R. A transparent, modular design supports reproducibility and flexibility across diverse portfolio sort configurations and adjustment methodologies. All code and data workflows are publicly available to promote standardization and reproducible research practices in empirical asset pricing.

Keywords: financial risk factors, reproducible research, asset pricing, API, R, Python.

1. Introduction: Constructing financial risk factors

In empirical finance, a key question is which sources of systematic risk drive asset returns. Researchers address this question by constructing factor returns: portfolios that isolate single economic characteristics or exposures. Factor returns serve model evaluation, hypothesis testing, benchmarking, and decomposition of returns into systematic and idiosyncratic components (Fama and French 1993; Cochrane 2011; Harvey, Liu, and Zhu 2016).

Despite their importance, no widely adopted, transparent, and accessible source standardizes factor construction. Current practice relies on ad-hoc publication of factor series, usually without code or detailed workflows, limiting reproducibility, comparability, and extension. Reproducible research requires transparent workflows, standardized data processing, analysis-

ready series, and modularity to vary methodological choices.[1]

Our project makes four contributions. First, we release a public factor library that centralizes series and workflows. Second, we provide a modular, end-to-end workflow for data acquisition, cleaning, variable construction, and portfolio aggregation. Third, we offer native R and Python implementations and a programmatic API for standardized data retrieval. Fourth, we publish all code, workflows, and documentation to enable replication, methodological comparison, and extensibility.

We provide two libraries, *tidyfinance* (on CRAN) and *py-tidyfinance*, that streamline data acquisition from WRDS, Compustat, and other commonly used data providers. Automated routines handle standard lagging procedures and eliminate ad-hoc fixes. The packages embed special adjustments typical for financial data, such as delistings and price or volume corrections.

A core focus of our project is portfolio sort construction, which involves many subjective choices: breakpoints, weighting, rebalancing, handling ties, and sample inclusion. Our code treats sorting as a configurable module. We document and test all choices and implement common variants (equal- or value-weighted, double sorts, quantile/percentile breaks, overlapping portfolios).

We illustrate the full workflow in R, from raw data to factor returns, including diagnostics. Python examples in the appendix ensure parity in configuration, defaults, and outputs.

We publish a curated factor library with preprocessed time series across sorting variables, weighting schemes, and adjustments. We demonstrate how methodological choices affect factor returns to highlight their empirical relevance. A programmatic API provides access to standardized series and preprocessed panels, with support for versioning and batch retrievals. We publish all software, workflows, and documentation to enable replication, sensitivity analysis, and extension.

2. Vignette: A case study in factor construction

Our core focus is to present the R package `tidyfinance`. We walk through the construction of a single risk factor, asset growth, from raw data to long-short portfolio returns. Each step generalizes to the full library of sorting variables. Figure 1 shows the end-to-end pipeline and highlights the `tidyfinance` functions involved at each stage.



Figure 1: Research workflow for factor construction with `tidyfinance`.

You can install the released version of `tidyfinance` from CRAN via:

[1] Existing resources illustrate the gap: the French Data Library (Fama and French 2023) provides canonical factors but no code; the Open Source Asset Pricing Project (Chen and Zimmermann 2022) focuses on replication; the global factor database (Jensen, Kelly, and Pedersen 2023) targets Python only. A modular, multilingual framework is still needed.

```
install.packages("tidyfinance")
```

You can install the development version of `tidyfinance` from GitHub via:

```
# install.packages("pak")
pak::pak("tidy-finance/r-tidyfinance")
```

Once installed, load the `tidyfinance` package.

```
library(tidyfinance)
```

For the sake of illustrating the functionalities of `tidyfinance`, we load a minimal set of additional libraries for data manipulation and visualization.

```
library(dplyr)
library(tidyr)
library(lubridate)
library(ggplot2)
```

2.1. Data sources

A central design goal of `tidyfinance` is to provide a single, consistent interface, `download_data()`, to a wide range of financial and economic data sources. The vignette only requires two of them (CRSP monthly stock returns and Compustat annual financial statements), but the same function gives access to 319 dataset types spanning equity returns, fixed-income transactions, macroeconomic predictors, and more. The package serves researchers well beyond asset pricing: macro forecasting (FRED, Goyal-Welch), credit markets (FISD, TRACE), and portfolio benchmarking (index constituents) all use the same unified workflow.

For our asset growth example, we download and preprocess the two required datasets as follows. Before the first WRDS download, users call `set_wrds_credentials()` to store their WRDS username and password securely in `.Renviron`; subsequent `download_data()` calls then establish the database connection automatically.

```
date_start <- as.Date("1960-01-01")
date_end <- as.Date("2024-12-31")
```

First, we download data from the Center for Research in Security Prices (CRSP). CRSP is a comprehensive database of historical stock market data, including prices, returns, and corporate actions for US equities. It is widely used in academic research for asset pricing, portfolio management, and financial econometrics.

```
crsp_monthly <- download_data(
  domain = "wrds",
  dataset = "crsp_monthly",
  version = "v2",
  start_date = date_start,
  end_date = date_end,
  add_ccm_links = TRUE
)
```

The CRSP dataset includes monthly stock returns, market capitalization, exchange codes, industry classifications, and excess returns (returns adjusted for the risk-free rate). The `add_ccm_links = TRUE` argument merges CRSP-Compustat linking information, which is essential for merging stock returns with financial statement data in later steps. The entire dataset contains 3451594 stock-month observations for 26263 unique stocks, covering the period from 1960-01-01 to 2024-12-01.

Next, we download data from Compustat, a comprehensive database of financial statements for publicly traded companies. Compustat provides both annual and quarterly data, but for this example, we only use the annual dataset.

```
compustat_annual <- download_data(
  domain = "wrds",
  dataset = "compustat_annual",
  start_date = date_start,
  end_date = date_end,
  additional_columns = c("at", "ib"),
  only_us = TRUE
)
```

Annual Compustat data contains 506782 firm-year observations for 39575 unique firms, covering the period from 1960-01-01 to 2024-12-01. The `additional_columns` argument specifies that we want to retrieve total assets (`at`) and income before extraordinary items (`ib`), which are necessary for computing asset growth and applying an earnings filter in later steps. Setting `only_us = TRUE` restricts the dataset to USD-denominated observations, ensuring consistency with the US stock returns from CRSP.

Behind the scenes, `download_data()` performs substantial cleaning and variable construction. For CRSP monthly data (version 2), the function restricts the stock universe to common equity securities of US-incorporated corporate issuers traded on NYSE, AMEX, or NASDAQ. The function computes market capitalization as shares outstanding times price (in millions), sets zero values to missing, and creates a one-month-lagged market capitalization for portfolio weights. Exchange codes map to human-readable labels and four-digit SIC codes map to broad industry groups (e.g., Manufacturing, Finance, Utilities). The function defines excess returns as raw returns minus the Fama-French risk-free rate, and drops rows with missing excess returns or market capitalization. CRSP version 2 already incorporates delisting returns into the return series, eliminating the need for a separate delisting bias adjustment.

For Compustat annual data, `download_data()` restricts to industrial-format, standardized, consolidated financial statements. The function constructs book equity via stockholders' equity (if this is not available, common equity plus preferred stock, and, as a last resort the total assets minus total liabilities), adjusts for deferred taxes and investment tax credits, and subtracts preferred stock (resolved through a coalesce hierarchy of redemption, liquidating, and par values). Negative or zero book equity values become missing. The function also computes operating profitability and investment ratios from standard Compustat variables, and retains only the most recent observation per firm-year to handle duplicate filings.

`py-tidyfinance` provides the same `download_data()` interface in Python, so researchers can use their preferred computational environment without sacrificing functionality.

Table 1 provides a complete overview of all data sources accessible via `download_data()` in both `tidyfinance` and `py-tidyfinance`.

Source	Description	Example
Ken French's Data Library	Fama-French factor models, portfolio returns, and industry classifications (169+ datasets at daily/weekly/monthly frequencies)	`download_data("famafrench", "Fama/French 5 Factors (2x3) [Daily]")`
Global Q	Hou-Xue-Zhang Q5 factor model (market, size, investment, ROE, expected growth) at multiple frequencies	`download_data("globalq", "factors_q5_monthly")`
Goyal-Welch Macro Predictors	Macroeconomic predictors: dividend-price ratio, earnings-price ratio, book-to-market, term spread, default yield spread, T-bill rate, inflation	`download_data("macro_predictors", "monthly")`
FRED	Any economic time series from the St. Louis Fed (GDP, CPI, unemployment, etc.)	`download_data("fred", series = c("GDP", "CPIAUCNS"))`
Yahoo Finance	Daily OHLCV stock price data with adjusted close for any ticker	`download_data("stock_prices", symbols = c("AAPL", "MSFT"))`
Index Constituents	Current constituents for 21 major indexes (S&P 500, Russell 2000, DAX, MSCI World, etc.)	`download_data("constituents", index = "DAX")`
Open Source Asset Pricing	213+ monthly asset pricing signals and anomaly variables	`download_data("osap")`
WRDS – CRSP	Monthly and daily stock returns, market cap, exchange, industry, excess returns	`download_data("wrds", "crsp_monthly")`
WRDS – Compustat	Annual and quarterly financial statements with customizable columns	`download_data("wrds", "compustat_annual")`
WRDS – CCM Links	CRSP-Compustat linking table mapping permno to gvkey	`download_data("wrds", "ccm_links")`
WRDS – FISD	Mergent Fixed Income Securities Database with bond characteristics	`download_data("wrds", "fisd")`

Source	Description	Example
WRDS – TRACE	Enhanced TRACE corporate bond transactions with Dick-Nielsen cleaning	`download_data("wrds", "trace_enhanced", cusips = "655844CJ5")`

Table 1: Data sources available via `download_data()` in `tidyfinance` and `py-tidyfinance`.

For programmatic discovery, `list_supported_types()` returns all supported datasets as a tibble, making it straightforward to explore available data without leaving the console.

```
list_supported_types() |> head()
```

```
# A tibble: 6 x 3
  type                 dataset_name               domain
  <chr>                <chr>                      <chr>
1 factors_q5_daily     q5_factors_daily_2024      Global Q
2 factors_q5_weekly    q5_factors_weekly_2024     Global Q
3 factors_q5_weekly_w2w q5_factors_weekly_w2w_2024 Global Q
4 factors_q5_monthly   q5_factors_monthly_2024    Global Q
5 factors_q5_quarterly q5_factors_quarterly_2024  Global Q
6 factors_q5_annual    q5_factors_annual_2024     Global Q
```

Similarly, `list_supported_indexes()` lists the 20 stock market indexes whose current constituents can be downloaded. Researchers can use these, for instance, to identify the member firms of a benchmark index before constructing portfolios.

```
list_supported_indexes() |> pull(index) |> head()
```

```
[1] "DAX"                          "EURO STOXX 50"
[3] "Dow Jones Industrial Average" "Russell 1000"
[5] "Russell 2000"                 "Russell 3000"
```

2.2. Data cleaning and variable construction

We illustrate data cleaning and variable construction using asset growth (AG) as our running example. Asset growth is the year-over-year percentage change in total assets (Cooper, Gulen, and Schill 2008),

$$AG_t = (AT_t - AT_{t-1})/AT_{t-1}.$$

The `add_lagged_columns()` function from `tidyfinance` handles temporal alignment by creating a lagged version of total assets grouped by firm identifier (`gvkey`) with a one-year lag. Non-finite values (arising when lagged total assets are zero or missing) become `NA`.[2]

[2]For the sake of simplicity we do not show every step of the data cleaning process in the paper. For instance, we apply an earnings filter to exclude firms with negative or zero income before extraordinary items (`ib`), which is a common practice in the literature to mitigate the influence of financial distress on asset growth portfolios. This filter is implemented in the code but not shown in the paper for brevity. We also apply an age filter to exclude firms younger than two years, which is common in the literature to ensure that the sorting variable is

```
sorting_variable_data <- compustat_annual |>
  add_lagged_columns(
    "at",
    lag = years(1),
    by = "gvkey"
  ) |>
  rename(at_lag1 = at_lag) |>
  mutate(
    asset_growth = (at - at_lag1) / at_lag1,
    asset_growth = if_else(is.finite(asset_growth), asset_growth, NA_real_)
  ) |>
  select(gvkey, date, asset_growth)
```

The next step merges annual sorting variables onto the monthly CRSP stock panel using `join_lagged_values()`. The function aligns fiscal-year-end data with monthly observations through a configurable lag window. Setting `ff_adjustment = TRUE` implements the Fama-French convention (Fama and French 1993), aligning fiscal-year data from year t with portfolio formation in July of year $t+1$ through a 12-to-18-month look-back window. The same lagging process generalizes to all sorting variables in the full factor library. The factor library API provides precomputed lagged panels for all variables.

Figure 2 contrasts the two lagging mechanisms. The left column uses a standard look-back window controlled by `lag` and `max_lag`. The right column implements the Fama-French convention (Fama and French 1993): fiscal-year data from year t becomes available only from July of year $t+1$, so all months from July through June share the same lagged value locked in at the previous rebalancing. In both cases, `add_lagged_columns()` selects the most recent observed value within the window.

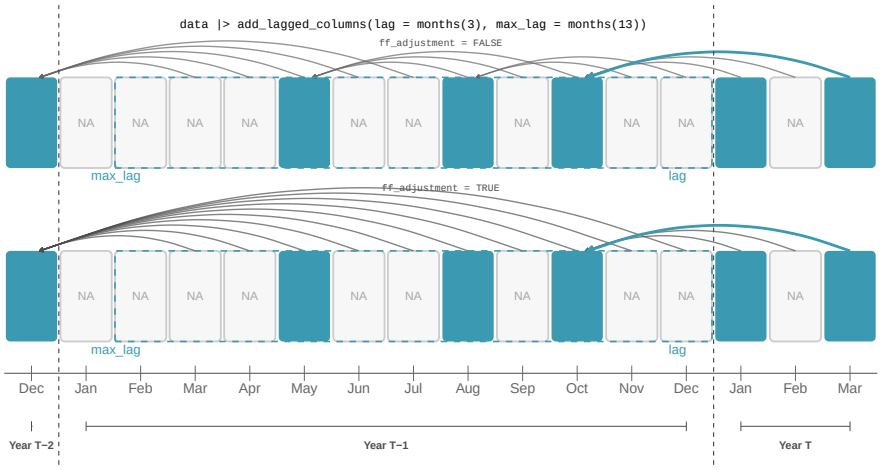


Figure 2: Two lagging mechanisms in `add_lagged_columns()`. Each column represents a data series with observed (filled) and missing (hollow) cells. Dashed rectangles mark the search window; arrows show which value gets matched to each target month.

based on stable financial information. This filter is also implemented in the code but not shown in the paper for brevity. The factor library API provides preprocessed panels with all these filters applied, so users can access the final sorting variables without having to implement these steps themselves.

```
sorting_data <- crsp_monthly |>
  join_lagged_values(
    sorting_variable_data,
    "gvkey",
    min_lag = months(12),
    max_lag = months(18),
    ff_adjustment = TRUE
  )

sorting_data <- sorting_data |>
  select(date, permno, exchange, ret_excess, mktcap_lag, asset_growth)
```

The resulting dataset merges monthly stock returns with the lagged sorting variable. Each row represents a stock-month observation, containing the information needed for portfolio construction: excess returns, exchange membership, industry classification, lagged market capitalization (for value-weighting), and the sorting variable itself.

Before turning to portfolio construction, we characterize the distribution of asset growth using `sorting_variable_data`, which contains one observation per firm-year and is the natural unit for cross-sectional analysis of annual accounting variables. Because asset growth is an annual characteristic that each firm contributes once per fiscal year, computing summary statistics at this level avoids the artificial replication that arises when the same value is carried forward across multiple months in `sorting_data`.

Asset growth is heavily right-skewed: a small number of firms with dramatic balance sheet expansions (through mergers, acquisitions, or large equity issuances) generate extreme positive values that dominate the mean. We therefore winsorize at the 1st and 99th percentiles before computing location and dispersion statistics. Table 2 reports the resulting pooled cross-sectional summary.

```
sorting_variable_data |>
  filter(!is.na(asset_growth)) |>
  mutate(
    asset_growth = pmax(
      pmin(asset_growth, quantile(asset_growth, 0.99)),
      quantile(asset_growth, 0.01)
    )
  ) |>
  summarise(
    N = n(),
    Mean = mean(asset_growth),
    SD = sd(asset_growth),
    P10 = quantile(asset_growth, 0.10),
    P25 = quantile(asset_growth, 0.25),
    Median = median(asset_growth),
    P75 = quantile(asset_growth, 0.75),
    P90 = quantile(asset_growth, 0.90)
  ) |>
  knitr::kable(digits = 3)
```

N	Mean	SD	P10	P25	Median	P75	P90
389124	0.238	0.858	-0.191	-0.034	0.064	0.207	0.588

Table 2: Cross-sectional summary statistics for asset growth (winsorized at the 1st and 99th percentiles). The table reports the number of firm-year observations (N), the mean, standard deviation (SD), and selected percentiles of the annual asset growth distribution computed from `sorting_variable_data`.

Table 3 complements the pooled view by reporting how cross-sectional moments evolve over time. For each calendar year we compute the cross-sectional mean, standard deviation, and median of (winsorized) asset growth, and then summarise those annual statistics across the full sample period.

```
sorting_variable_data |>
  filter(!is.na(asset_growth)) |>
  mutate(year = lubridate::year(date)) |>
  group_by(year) |>
  mutate(
    p01 = quantile(asset_growth, 0.01),
    p99 = quantile(asset_growth, 0.99),
    asset_growth = pmax(pmin(asset_growth, p99), p01)
  ) |>
  summarise(
    N = n(),
    Mean = mean(asset_growth),
    SD = sd(asset_growth),
    Median = median(asset_growth),
    .groups = "drop"
  ) |>
  summarise(
    Years = n(),
    `Avg N` = round(mean(N)),
    `Mean (avg)` = mean(Mean),
    `Mean (min)` = min(Mean),
    `Mean (max)` = max(Mean),
    `SD (avg)` = mean(SD),
    `Median (avg)` = mean(Median),
    `Median (min)` = min(Median),
    `Median (max)` = max(Median)
  ) |>
  knitr::kable(digits = 3)
```

Years	Avg N	Mean (avg)	Mean (min)	Mean (max)	SD (avg)	Median (avg)	Median (min)	Median (max)
64	6080	0.242	0.048	0.889	0.847	0.067	0.001	0.132

Years	Avg N	Mean (avg)	Mean (min)	Mean (max)	SD (avg)	Median (avg)	Median (min)	Median (max)

Table 3: Time-series averages of annual cross-sectional summary statistics for asset growth (winsorized at the 1st and 99th percentiles). For each calendar year, cross-sectional mean, standard deviation, and median are computed from `sorting_variable_data`; the table reports the time-series mean, minimum, and maximum of those annual statistics.

Having characterised the distribution of the sorting variable, we now turn to `sorting_data`, the firm-month panel that serves as direct input to portfolio construction. Figure 3 shows the 33rd, 50th, and 66th percentiles of asset growth over time. Pronounced spikes during economic expansions and contractions highlight the cyclical nature of corporate investment that portfolio sorts aim to exploit.

```
sorting_data |>
  group_by(date) |>
  summarise(
    p33 = quantile(asset_growth, 0.33, na.rm = TRUE),
    p50 = quantile(asset_growth, 0.50, na.rm = TRUE),
    p66 = quantile(asset_growth, 0.66, na.rm = TRUE),
    .groups = "drop"
  ) |>
  pivot_longer(
    cols = c(p33, p50, p66),
    names_to = "quantile",
    values_to = "asset_growth"
  ) |>
  ggplot(aes(x = date, y = asset_growth, color = quantile)) +
  geom_line() +
  labs(x = NULL, y = "Asset growth", color = "Quantile") +
  theme(legend.position = "bottom")
```

```
Warning: Removed 72 rows containing missing values or values outside the scale range
(`geom_line()`).
```

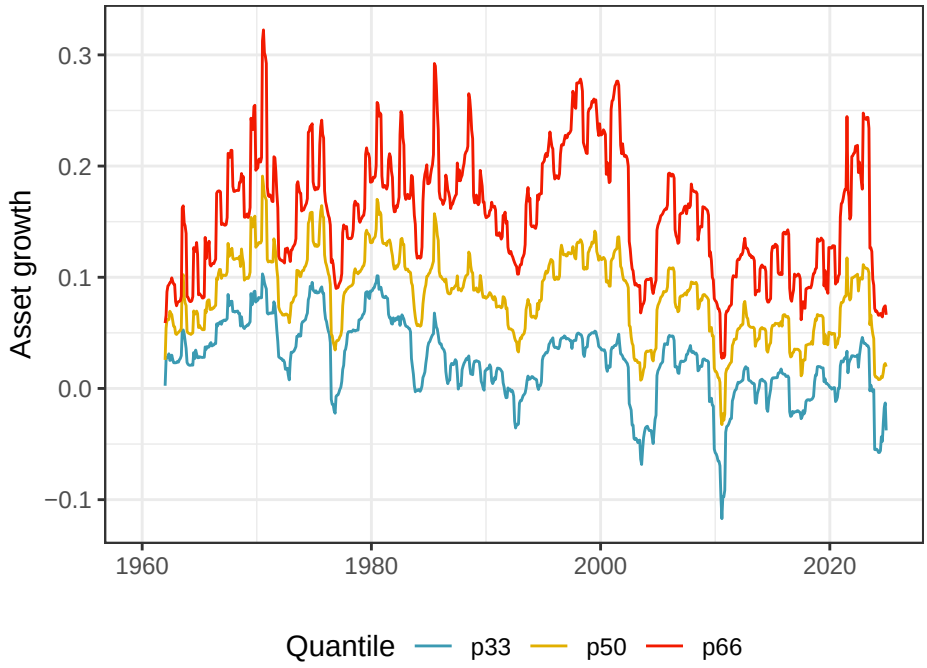


Figure 3: Distribution of asset growth over time. Lines show the 33rd, 50th, and 66th percentiles of the cross-sectional distribution of asset growth in `sorting_data`, the firm-month panel used as input for portfolio construction.

With the sorting variable constructed and merged onto the monthly stock panel, we now turn to portfolio sorts, the mechanism that translates firm-level characteristics into tradable factor returns.

2.3. Portfolio sorts

In portfolio sorts, stocks are ranked each period by their sorting variable and divided into quantile portfolios. Portfolio returns are equal-weighted or value-weighted averages of constituent stock returns. The long-short factor return is the difference between the extreme portfolios, for instance bottom minus top for asset growth, reflecting the negative premium documented by Cooper *et al.* (2008). Key methodological decisions include the number of portfolios, the exchange universe for computing breakpoints, the weighting scheme, and the rebalancing frequency.

The `tidyfinance` package implements this workflow through three functions. First, `breakpoint_options()` configures the sorting parameters: the number of quantile portfolios and which exchanges determine the breakpoints. Second, `compute_portfolio_returns()` takes the sorting data, applies the specified configuration, and returns portfolio-level returns. Third, `compute_long_short_returns()` constructs the hedge portfolio return from the extreme quantiles.

Univariate sorts

TODO: Discuss the subfunctions from within `compute_portfolio_returns()`, such as `tidyfinance::assign_portfolios()`, which assigns each stock-month observation to a quantile portfolio based on the breakpoints, and `?tidyfinance::compute_breakpoints()`, which computes the quantile breakpoints for the sorting variable based on the specified exchange universe. These functions are modular and can be used independently for more customized portfolio constructions, such as overlapping portfolios or alternative weighting schemes.

In a univariate sort, stocks are ranked solely on a single characteristic. We illustrate this with asset growth, forming decile portfolios using NYSE breakpoints and value-weighted returns, rebalanced annually in July.

```
options_main <- breakpoint_options(
  n_portfolios = 10,
  breakpoint_exchanges = "NYSE"
)
```

The `breakpoint_options()` call specifies that breakpoints come from NYSE-listed stocks only, a common convention that avoids the influence of small AMEX and NASDAQ firms on portfolio cutoffs. We then pass these options to `compute_portfolio_returns()`.

```
portfolio_returns <- compute_portfolio_returns(
  sorting_data,
  sorting_variables = "asset_growth",
  sorting_method = "univariate",
  rebalancing_month = 7,
  options_main
)
```

```
Returning a complete panel with 301 missing values in factor returns due to
insufficient observations (n() < 0).
```

`compute_portfolio_returns()` returns a long tibble with one row per portfolio-month, containing value-weighted (`ret_excess_vw`), equal-weighted (`ret_excess_ew`), and capped value-weighted (`ret_excess_vw_capped`) excess returns. The long-short factor return is obtained via `compute_long_short_returns()`. The `direction = "bottom_minus_top"` argument means the factor is long low-AG stocks and short high-AG stocks, consistent with the negative premium documented by Cooper *et al.* (2008).

```
factor_returns_ag <- portfolio_returns |>
  compute_long_short_returns(direction = "bottom_minus_top")
```

Table 4 summarises the time series of each decile across all three weighting schemes, with the long-short portfolio appended as the final row. A monotone decline from low- to high-AG portfolios is consistent with the negative asset growth premium.

```
decile_stats <- portfolio_returns |>
  drop_na() |>
  group_by(portfolio) |>
```

```
  summarise(
    vw_mean   = mean(ret_excess_vw) * 12,
    vw_tstat  = mean(ret_excess_vw) / (sd(ret_excess_vw) / sqrt(n())),
    ew_mean   = mean(ret_excess_ew) * 12,
    ew_tstat  = mean(ret_excess_ew) / (sd(ret_excess_ew) / sqrt(n())),
    vwc_mean  = mean(ret_excess_vw_capped) * 12,
    vwc_tstat = mean(ret_excess_vw_capped) / (sd(ret_excess_vw_capped) / sqrt(n())),
    .groups = "drop"
  ) |>
  mutate(portfolio = as.character(portfolio))

ls_row <- factor_returns_ag |>
    drop_na() |>
  summarise(
    portfolio = "L/S",
    vw_mean   = mean(ret_excess_vw) * 12,
    vw_tstat  = mean(ret_excess_vw) / (sd(ret_excess_vw) / sqrt(n())),
    ew_mean   = mean(ret_excess_ew) * 12,
    ew_tstat  = mean(ret_excess_ew) / (sd(ret_excess_ew) / sqrt(n())),
    vwc_mean  = mean(ret_excess_vw_capped) * 12,
    vwc_tstat = mean(ret_excess_vw_capped) / (sd(ret_excess_vw_capped) / sqrt(n()))
  )

bind_rows(decile_stats, ls_row) |>
  knitr::kable(
    digits = 3,
    col.names = c("Portfolio", "VW Mean", "VW t-stat", "EW Mean", "EW t-
stat", "VWC Mean", "VWC t-stat")
  )
```

Portfolio	VW Mean	VW t-stat	EW Mean	EW t-stat	VWC Mean	VWC t-stat
1	0.131	3.945	0.157	3.830	0.111	3.211
2	0.088	3.329	0.141	4.409	0.099	3.598
3	0.150	5.422	0.144	5.405	0.141	5.520
4	0.135	5.344	0.142	5.286	0.127	5.199
5	0.105	4.107	0.117	4.416	0.113	4.356
6	0.095	3.738	0.137	4.291	0.107	4.066
7	0.082	2.877	0.125	4.446	0.102	3.777
8	0.118	4.226	0.124	4.377	0.123	4.455
9	0.127	4.229	0.099	3.324	0.103	3.502
10	0.116	3.358	0.058	1.686	0.062	1.859
L/S	0.016	0.568	0.099	3.483	0.049	2.152

	VW		EW		VWC	
Portfolio	Mean	VW t-stat	Mean	EW t-stat	Mean	VWC t-stat

Table 4: Time-series averages of annualised monthly excess returns for asset growth decile portfolios. Portfolios are formed annually in July using NYSE breakpoints, rebalanced annually. Means are annualised (multiplied by 12). The t-statistic tests whether the mean monthly excess return differs from zero. The L/S row is the long-short portfolio (bottom minus top decile).

Bivariate sorts

Bivariate (double) sorts control for a second variable, typically firm size (`mktcap_lag`), when forming portfolios. In dependent sorts, stocks are first sorted into groups by the secondary variable; within each group, they are then sorted on the main characteristic. In independent sorts, both sorts apply unconditionally, and portfolios form at the intersections. The factor return averages across the secondary dimension, isolating the effect of the primary variable after controlling for size.

```
options_main <- breakpoint_options(
  n_portfolios = 5,
  breakpoint_exchanges = "NYSE"
)
options_secondary <- breakpoint_options(
  n_portfolios = 2,
  breakpoint_exchanges = "NYSE"
)

portfolio_returns_biv <- compute_portfolio_returns(
  sorting_data,
  c("asset_growth", "mktcap_lag"),
  sorting_method = "bivariate-dependent",
  rebalancing_month = 7,
  options_main,
  options_secondary
)
```

```
Returning a complete panel with 150 missing values in factor returns due to
insufficient observations (n() < 0).
```

```
factor_returns_biv <- portfolio_returns_biv |>
  compute_long_short_returns(direction = "bottom_minus_top")

factor_returns_biv |> head()
```

```
# A tibble: 6 x 3
  date       ret_excess_vw ret_excess_ew
  <date>             <dbl>         <dbl>
```

Preprocessing choice	Argument	Options
Size exclusion	‘exclude_size‘	None; *bottom 20%*
Exclusion of financials	‘exclude_financials‘	*No*; Yes
Exclusion of utilities	‘exclude_utilities‘	*No*; Yes
Exclusion of neg. earnings	‘exclude_negative_earnings‘	*No*; Yes
Sorting variable lag	‘sorting_variable_lag‘	3m; *6m*; FF
Rebalancing	‘rebalancing‘	*Monthly*; Annual
BP: Quantiles (main)	‘breakpoints_main‘	3; 5; *10*
Sort method	‘sorting_method‘	*Single*; Double dependent; Double independ
BP: Quantiles (secondary)	‘breakpoints_secondary‘	*NA*; 2; 5
BP: Exchanges	‘breakpoints_exchanges‘	*NYSE*; All
Weighting scheme	‘weighting_scheme‘	EW; *VW*; truncated VW

Table 5: Preprocessing choices in the portfolio sort grid and corresponding `download_data()` arguments. Italic values indicate the default. For double sorts, the secondary quantiles and exchange choices apply to the size dimension only; for univariate sorts, those rows are not applicable. FF refers to the lag procedure of Fama and French (1993).

```
1 1960-01-01            NA            NA
2 1960-02-01            NA            NA
3 1960-03-01            NA            NA
4 1960-04-01            NA            NA
5 1960-05-01            NA            NA
6 1960-06-01            NA            NA
```

Dependent sorts are appropriate when the secondary variable (here, size) strongly affects the distribution of the primary sorting variable, as is typical for many accounting-based characteristics. Independent sorts treat both dimensions symmetrically and work best when both variables are of equal conceptual interest. The `sorting_method` argument supports both methods (`"bivariate-dependent"` or `"bivariate-independent"`).

3. Factor library and API

Specification grid and the factor library

The examples above represent single points in a high-dimensional space of methodological choices. Table 5 summarises the preprocessing decisions that can be varied when constructing factor returns. The full factor library exhaustively evaluates all valid combinations of these choices for each of the sorting variables listed in Table 6.

We host precomputed factor returns for all specifications on Hugging Face and make them accessible via the `tidyfinance` API. Researchers can retrieve results without re-running the computationally intensive portfolio sort pipeline. An interactive download interface is also available via the Shiny app at https://app-download-center.cloud.sdu.dk/.

Section 3 provides a comprehensive overview of all available sorting variables and their empirical properties.

The factor library covers 50 sorting variables spanning seven major categories: investment, profitability, valuation, momentum, intangibles, trading frictions, and financing. The selection follows the cross-section of characteristics studied by Jensen *et al.* (2023), focusing on variables with established empirical support in the asset pricing literature. Table 6 lists all sorting variables, their long-short direction, and the primary reference for each. For each variable, the full factor library provides precomputed factor returns across all specification paths defined in Table 5. We host the complete dataset on Hugging Face at https://huggingface.co/datasets/tidy-finance/factor-library. Researchers can also browse and retrieve factor returns through an interactive Shiny app at https://app-download-center.cloud.sdu.dk/.

Variable	Name		Main reference
52w	52-Week High	↑	George and Hwang (2004)
aci	Abnormal Corporate Investment	↓	Titman, Wei, and Xie (2004)
ami	Amihud Illiquidity	↑	Amihud (2002)
ag	Asset Growth	↓	Cooper *et al.* (2008)
beta	Beta	↑	Fama and MacBeth (1973)
bm	Book-to-Market	↑	Daniel and Titman (2006)
cfm	Cash Flow to Market	↑	Lakonishok, Shleifer, and Vishny (1994)
cbop	Cash-Based Operating Profitability	↑	Ball, Gerakos, Linnainmaa, and Nikolaev (2016)
dbe	Change in Common Equity	↓	Richardson, Sloan, Soliman, and Tuna (2005)
dcol	Change in Current Operating Liabilities	↓	Richardson *et al.* (2005)
dfnl	Change in Financial Liabilities	↓	Richardson *et al.* (2005)
dnoa	Change in Net Operating Assets	↓	Hirshleifer, Hou, Teoh, and Zhang (2004)
dpia	Change in PPE and Intangibles	↓	Lyandres, Sun, and Zhang (2008)
dwc	Change in Working Capital	↓	Richardson *et al.* (2005)
cdi	Composite Debt Issuance	↓	Lyandres *et al.* (2008)
csi	Composite Share Issuance	↓	Daniel and Titman (2006)
dm	Debt to Market	↑	Bhandari (1988)
dtv	Dollar Trading Volume	↓	Brennan, Chordia, and Subrahmanyam (1998)
em	Earnings to Market	↑	Basu (1983)
ebm	Enterprise Book to Market	↑	Penman, Richardson, and Tuna (2007)
gpa	Gross Profits to Assets	↑	Novy-Marx (2013)
hr	Hiring Rate	↓	Belo and Lin (2012)
iskew	Idiosyncratic Skewness	↓	Bali, Engle, and Murray (2016)
ivol	Idiosyncratic Volatility	↓	Ang, Hodrick, Xing, and Zhang (2006)
dinv	Inventory Changes	↓	Thomas and Zhang (2002)
ig	Investment Growth	↓	Xing (2008)

rev	Long-Term Reversal	↓	De Bondt and Thaler (1985)
me	Market Equity	↓	Banz (1981)
mdr	Maximum Daily Return	↓	Bali, Cakici, and Whitelaw (2011)
mom	Momentum	↑	Fama and French (1996)
ndf	Net Debt Financing	↓	Baker and Wurgler (2006)
ndm	Net Debt to Market	↓	Penman *et al.* (2007)
nef	Net Equity Financing	↓	Baker and Wurgler (2006)
nxf	Net External Financing	↓	Baker and Wurgler (2006)
noa	Net Operating Assets	↓	Hirshleifer *et al.* (2004)
npy	Net Payout Yield	↑	Boudoukh, Michaely, Richardson, and Roberts (2007)
o	O-Score	↓	Ohlson (1980); Dichev (1998)
oa	Operating Accruals	↓	Sloan (1996)
ocm	Operating Cash Flow to Market	↑	Desai, Rajgopal, and Venkatachalam (2004)
ope	Operating Efficiency	↑	Fama and French (2015)
ol	Operating Leverage	↑	Novy-Marx (2011)
rer	Real-Estate Ratio	↑	Titman *et al.* (2004)
rs	Relative Strength	↑	Jegadeesh and Livnat (2006)
rmom	Residual Momentum	↑	Blitz, Huij, and Martens (2011)
roa	Return on Assets	↑	Balakrishnan, Bartov, and Faurel (2010)
roe	Return on Equity	↑	Hou, Xue, and Zhang (2014)
sm	Sales to Market	↑	Barbee Jr, Mukherji, and Raines (1996)
tur	Share Turnover	↓	Datar, Naik, and Radcliffe (1998)
sue	Standardized Unexpected Earnings	↑	Foster, Olsen, and Shevlin (1984)
tbi	Taxable Income to Book Income	↓	Lev and Nissim (2004)

Table 6: Sorting variables in the factor library. The direction column indicates whether the long-short factor return goes long top-minus-bottom (↑) or bottom-minus-top (↓) of the characteristic distribution.

To download factor returns from the factor library, simply call

```
asset_growth_returns <- download_data("tidyfinance", "factor_library",
sorting_variable = "ag", sorting_variable_lag = "6m")
```

Note on `dataset = "factor_library"` defaults: The defaults reflect one common portfolio construction choice, but may not suit every research question. Always verify that the selected combination matches your intended design.

```
download_data("tidyfinance", "factor_library",
  sorting_variable = "52w",
  sorting_variable_lag = "6m",
  fill_all = TRUE
)
```

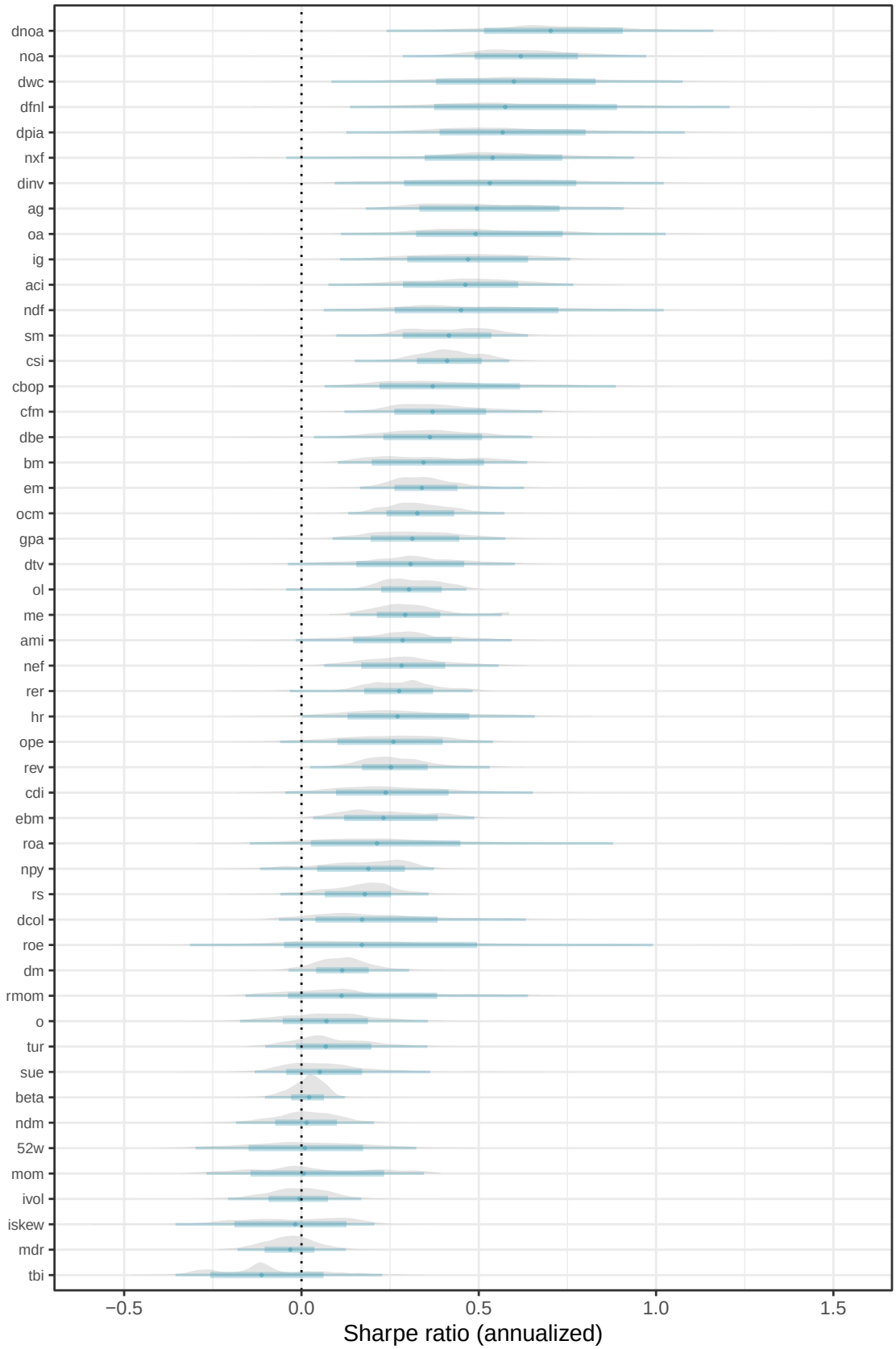


Figure 4: **Sharpe ratios.** The ridge plots indicate the distribution of annualized Sharpe ratios of long-short return differentials for a given sorting variable across all data preprocessing choices illustrated in Table 5. The estimates are based on the available sample period for each factor.

```
# A tibble: 1,927,680 x 16
      id date       ret_type      ret sorting_variable exclude_size
   <int> <date>     <chr>       <dbl> <chr>                    <dbl>
 1 62337 1962-04-01 vw         0.0214  52w                          0
 2 62337 1962-04-01 ew         0.0110  52w                          0
 3 62337 1962-05-01 vw         0.0219  52w                          0
 4 62337 1962-05-01 ew        -0.00353 52w                          0
 5 62337 1962-06-01 vw         0.0213  52w                          0
 6 62337 1962-06-01 ew         0.0111  52w                          0
 7 62337 1962-07-01 vw        -0.0496  52w                          0
 8 62337 1962-07-01 ew        -0.00689 52w                          0
 9 62337 1962-08-01 vw        -0.00269 52w                          0
10 62337 1962-08-01 ew        -0.0129  52w                          0
# i 1,927,670 more rows
# i 10 more variables: exclude_financials <lgl>, exclude_utilities <lgl>,
#   exclude_negative_earnings <lgl>, sorting_variable_lag <chr>,
#   rebalancing <chr>, breakpoints_main <dbl>, sorting_method <chr>,
#   breakpoints_secondary <dbl>, breakpoints_exchanges <chr>,
#   weighting_scheme <chr>
```

To provide specific requests for handselected ids, users can call

```
full_grid <- get_available_huggingface_files(
    "tidy-finance",
    "factor-library-grid"
  ) |>
    dplyr::pull(.data$url) |>
    arrow::read_parquet()

download_factor_library_returns_ids(c(123, 456))
```

4. Summary and discussion

Computational details

We obtained the results in this paper using R~4.5.2 with the following packages: **tidyfinance**~0.4.5.9021, **dplyr**~1.2.0, **tidyr**~1.3.1, **ggplot2**~4.0.0, **lubridate**~1.9.4, **arrow**~21.0.0.1, **stringr**~1.5.2, **forcats**~1.0.0, **ggdist**~3.3.3, and **knitr**~1.51. CRAN hosts R and all packages at https://CRAN.R-project.org/. `renv`~1.1.5 ensures full reproducibility of the computational environment.

We publish all code, data workflows, and documentation at https://github.com/tidy-finance/jss-multilingual-factor-library.

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Technical details

Internet Appendix

Variable construction

In this section, we show the construction details for the risk factors. We group them by their underlying mechanism as shown in Table 6 in the Appendix of the main paper.

IA-1. Financing

CDI - Composite Debt Issuance

Following Lyandres *et al.* (2008), we compute composite debt issuance as the logarithmic growth rate of book debt (i.e., current debt, DLC, plus long-term debt, DLTT) over the last five years, i.e.,

$$\text{CDI}_t = \log(\text{DLC}_t + \text{DLTT}_t) - \log(\text{DLC}_{t-60} + \text{DLTT}_{t-60})$$

We replace the missing values of DLC and DLTT with zero individually. Additionally, the sum of DLC and DLTT must be larger than zero, and is set to missing otherwise.

CSI - Composite Share Issuance

Following Daniel and Titman (2006), we measure composite share issuance from CRSP data as the difference between the change in market equity (ME) (i.e., the shares outstanding, shrout, times the absolute value of their price, altprc, in million USD) and the cumulative log return (where r is a simple return) in each month from t to $t-60$, i.e.,

$$\text{CSI}_t = \log(ME_t/ME_{t-60}) - \sum_{\tau=t-59}^{t} \log(1+r_\tau)$$

DBE - Change in Common Equity

Following Richardson *et al.* (2005), we calculate the change in common equity as the change in common equity (CEQ) over one year scaled by lagged assets (AT), i.e.,

$$DBE_t = \frac{CEQ_t - CEQ_{t-12}}{AT_{t-12}}$$

DCOL - Change in Current Operating Liabilities

Following Richardson *et al.* (2005), we measure the change in current operating liabilities as the change in current operating liabilities (i.e., current liabilities, LCT, minus current debt, DLC) over one year scaled by lagged assets (AT), i.e.,

$$DCOL_t = \frac{(LCT_t - DLC_t) - (LCT_{t-12} - DLC_{t-12})}{AT_{t-12}}$$

We replace the missing values of DLC with 0.

DFNL - Change in Financial Liabilities

Following Richardson *et al.* (2005), we measure the change in financial liabilities (i.e., the sum of current debt, DLC, plus long-term debt, DLTT, plus preferred stocks, PSTK) over one year scaled by lagged assets (AT), i.e.,

$$DFNL_t = \frac{(DLC_t + DLTT_t + PSTK_t) - (DLC_{t-12} + DLTT_{t-12} + PSTK_{t-12})}{AT_{t-12}}$$

We replace the missing values of DLC, DLTT and PSTK with 0, if at least one of them is given.

NDF - Net Debt Financing

Following Baker and Wurgler (2006), we measure net debt financing as cash proceeds from the issuance of long-term debt (DLTIS) minus cash payments for long-term debt reductions (DLTR) plus net changes in current debt (DLCCH) scaled by the average total assets (AT) over the current and the last year, i.e.,

$$NDF_t = \frac{DLTIS_t - DLTR_t + DLCCH_t}{\frac{1}{2}(AT_t + AT_{t-12})}$$

We replace the missing values of DLCCH with zero.

NEF - Net Equity Financing

Following Baker and Wurgler (2006), we measure net equity financing as proceeds from the sale of common and preferred stocks (SSTK) minus payments for the repurchase of common and preferred stocks (PRSTKC) minus cash payments for dividends (DV) scaled by the average total assets (AT) over the current and the last year, i.e.,

$$NEF_t = \frac{SSTK_t - PRSTKC_t - DV_t}{\frac{1}{2}(AT_t + AT_{t-12})}$$

NXF - Net External Financing

Following Baker and Wurgler (2006), we measure net external financing as the sum of net debt financing (NDF) and net equity financing (NEF) as defined above.

IA-2. Intangibles

HR - Hiring Rate

Following Belo and Lin (2012), we measure the hiring rate as the change in the number of employees (EMP) over one year scaled by the average number of employees, i.e.,

$$HR_t = \frac{EMP_t - EMP_{t-12}}{\frac{1}{2}(EMP_t + EMP_{t-12})}$$

We exclude firms with a hiring rate of zero.

OL - Operating Leverage

Following Novy-Marx (2011), we measure operating leverage as cost of goods sold (COGS) plus selling, general, and administrative expenses (XSGA), both scaled by current total assets (AT), i.e.,

$$OL_t = \frac{COGS_t + XSGA_t}{AT_t}$$

RER - Real-Estate Ratio

Following Titman *et al.* (2004), we measure the real-estate ratio in two subperiods differently.

(a) Prior to 1983, the real-estate ratio is the sum of buildings (PPENB) and capital leases (PPENLS) scaled by net property, plant, and equipment (PPENT), i.e.,

$$RER_t = \frac{PPENB_t + PPENLS_t}{PPENT_t}$$

(b) After the end of 1983, the real-estate ratio is the sum of buildings at cost (FATB) and leases at cost (FATL), both divided by gross property, plant, and equipment (PPEGT), i.e.,

$$RER_t = \frac{FATB_t + FATL_t}{PPEGT_t}$$

Subsequently, we winsorize the real-estate ratios in each fiscal year at the first and 99th percentile. The industry-adjusted real-estate ratio is obtained by subtracting the industry average real-estate ratio from each stock-specific real-estate ratio. We use 2-digit SIC codes to assign stocks to industries. We always require at least five observations to calculate the industry average each year.

IA-3. Investment

ACI - Abnormal Corporate Investment

Following Titman *et al.* (2004), we measure abnormal corporate investments using the variable CE, which is capital expenditures (CAPX) scaled by sales (SALE), then ACI is given by CE scaled by CE from the previous three years minus one, i.e.,

$$ACI_t = \frac{CE_t}{\frac{1}{3}(CE_{t-12} + CE_{t-24} + CE_{t-36})} - 1$$

We follow Hou, Xue, and Zhang (2020) and exclude stocks with sales below 10 million dollars.

AG - Asset Growth

Following Cooper *et al.* (2008), we measure asset growth as the change in total assets (AT) over one year scaled by lagged assets (AT), i.e.,

$$AG_t = \frac{AT_t - AT_{t-12}}{AT_{t-12}}$$

DINV - Inventory Changes

Following Thomas and Zhang (2002), we measure the change in inventory as the annual change in inventories (INVT) scaled by average total assets (AT) over the current and the last year, i.e.,

$$DINV_t = \frac{INVT_t - INVT_{t-12}}{\frac{1}{2}(AT_t + AT_{t-12})}$$

DNOA - Change in Net Operating Assets

Following Hirshleifer *et al.* (2004), we measure net operating assets (defined as NOA^{int}) as

$$NOA_t^{int} = (AT_t - CHE_t) - (AT_t - DLC_t - DLTT_t - MIB_t - PSTK_t - CEQ_t)$$

where AT corresponds to total assets, CHE to cash and short-term investments, DLC to current debt, DLTT to long-term debt, MIB to minority interests, PSTK to the value of preferred stocks, and CEQ to common equity. Missing values in DLC, DLTT, MIB, and PSTK are set to zero. Then, the change in net operating assets is the difference between net operating assets of fiscal year t and fiscal year $t-12$ scaled by lagged assets (AT), i.e.,

$$DNOA_t = \frac{NOA_t^{int} - NOA_{t-12}^{int}}{AT_{t-12}}$$

DPIA - Change in Property, Plant, Equipment, and Inventory

Following Lyandres *et al.* (2008), we measure the change in property, plant, equipment, and inventory by combining the change in gross property, plant, and equipment (PPEGT) with the annual change in inventory (INVT) and scaling this sum by lagged total assets (AT), i.e.,

$$DPIA_t = \frac{(PPEGT_t - PPEGT_{t-12}) + (INVT_t - INVT_{t-12})}{AT_{t-12}}$$

DWC - Change in Net Non-Cash Working Capital

Following Richardson *et al.* (2005), we measure non-cash working capital as

$$WC_t = (ACT_t - CHE_t) - (LCT_t - DLC_t)$$

where ACT corresponds to current assets, CHE to cash, LCT to current liabilities, and DLC to short-term debt. We set missing values of DLC to zero. Then, the change in net non-cash

working capital corresponds to the change in WC from fiscal year $t-12$ to fiscal year t scaled by lagged total assets (AT), i.e.,

$$DWC_t = \frac{WC_t - WC_{t-12}}{AT_{t-12}}$$

IG - Investment Growth

Following Xing (2008), we measure investment growth as the annual change in capital expenditures (CAPX) scaled by lagged capital expenditures, i.e.,

$$IG_t = \frac{CAPX_t - CAPX_{t-12}}{CAPX_{t-12}}$$

NOA - Net Operating Assets

Following Hirshleifer *et al.* (2004), we measure net operating assets (defined as NOA^{int}) as

$$NOA_t^{int} = (AT_t - CHE_t) - (AT_t - DLC_t - DLTT_t - MIB_t - PSTK_t - CEQ_t)$$

where AT corresponds to total assets, CHE to cash and short-term investments, DLC to current debt, DLTT to long-term debt, MIB to minority interests, PSTK to the value of preferred stocks, and CEQ to common equity. Missing values in DLC, DLTT, MIB, and PSTK are set to zero. Then, net operating assets is the net operating assets of fiscal year t scaled by lagged assets (AT), i.e.,

$$NOA_t = \frac{NOA_t^{int}}{AT_{t-12}}$$

OA - Operating Accruals

Following Sloan (1996) (before 1988) and Hribar and Collins (2002) (after 1988), we measure operating accruals until 1988 as

$$OA_t = \frac{(\Delta ACT_t - \Delta CHE_t) - (\Delta LCT_t - \Delta DLC_t - \Delta TXP_t) - DP_t}{AT_{t-12}}$$

where ACT is current assets, CHE cash, LCT current liabilities, DLC short-term debt, TXP taxes payable, and DP depreciation and amortization. Moreover, we replace the missing values of DLC and TXP with zero. Δ represents changes over one year. From 1988 onwards, we measure operating accruals as

$$OA_t = \frac{NI_t - OANCF_t}{AT_{t-12}}$$

where NI is net income and OANCF corresponds to net cash flow from operations.

IA-4. Momentum

MOM - Return Momentum

Following Fama and French (1996), we measure return momentum from monthly CRSP returns (where r is a simple return) as the cumulative return from month $t-11$ to month $t-1$ skipping the most recent month, i.e.,

$$\text{MOM}_t = \exp\left(\sum_{\tau=t-11}^{t-1} \log(1+r_\tau)\right) - 1$$

RMOM - Residual Momentum

Following Blitz *et al.* (2011), we define the 11-month residual momentum (RMOM) in each month and for each stock as cumulative residual returns from month $t-11$ to month $t-1$, scaled by the standard deviation of residual returns over the same time horizon. Residual returns ϵ are obtained in each month from regressing monthly simple, excess stock returns from month $t-1$ to month $t-36$ on the Fama and French (1993) three-factor model. Throughout these rolling regressions, we always require 36 monthly returns.

RS - Revenue Surprise

Following Jegadeesh and Livnat (2006), we measure the change in revenues per share (DRPS) from quarterly Compustat data as quarterly sales (SALEQ) scaled by the product of quarterly shares outstanding (CSHPRQ) times their cumulative adjustment factor (AJEXQ), i.e.,

$$DRPS_q = \frac{SALEQ_q}{CSHPRQ_q \cdot AJEXQ_q} - \frac{SALEQ_{q-4}}{CSHPRQ_{q-4} \cdot AJEXQ_{q-4}}$$

Then, the revenue surprise corresponds to the change in revenues per share over the last four quarters scaled by the standard deviation of the change in revenues per share over the last eight quarters. We require at least six quarterly observations for this rolling standard deviation and remove infinite values. Finally, the earnings announcement date has to be after the fiscal quarter end date.

SUE - Standardized Unexpected Earnings

Following Foster *et al.* (1984), we measure unexpected earnings per share (UES) from quarterly Compustat data as the change in split-adjusted earnings (which are equal to quarterly earnings per share, EPSPXQ, scaled by the correction factor for shares outstanding, AJEXQ) per share from its value four quarters ago

$$UES_q = \frac{EPSPXQ_q}{AJEXQ_q} - \frac{EPSPXQ_{q-4}}{AJEXQ_{q-4}}$$

Then, standardized unexpected earnings are defined as unexpected earnings per share divided by the standard deviation of unexpected earnings per share over the previous eight quarters. We require at least six quarterly observations for this rolling standard deviation. Finally, the earnings announcement date has to be after the fiscal quarter end date.

52W - 52-Week High

Following George and Hwang (2004), we measure the 52-week high as month t's closing split-adjusted stock price scaled by the highest daily split-adjusted stock price over the previous twelve months. We only consider stock-month pairs with more than three valid price observations.

REV - Long-Term Reversal

Following De Bondt and Thaler (1985), we measure the long-term reversal effect by the cumulative returns from month $t-60$ to month $t-13$, i.e.,

$$\text{REV}_t = \exp\left(\sum_{\tau=t-60}^{t-13} \log(1+r_\tau)\right) - 1$$

IA-5. Profitability

CBOP - Cash-Based Operating Profitability

Following Ball *et al.* (2016), we measure cash-based operating profitability

$$\begin{aligned} CBOP_t =& (REVT_t - COGS_t - XSGA_t + XRD_t \\ & - \Delta RECT_t - \Delta INVT_t - \Delta XPP_t + \Delta DRC_t \\ & + \Delta DRLT_t + \Delta AP_t + \Delta XACC_t)/AT_t \end{aligned}$$

where REVT is total revenue, COGS are cost of goods sold, XSGA are selling, general, and administrative expenses, and XRD are R&D expenses. Moreover, Δ refers to one-year changes, which we measure for accounts receivable (RECT), inventory (INVT), prepaid expenses (XPP), deferred revenues (the sum of current, DRC, and long-term, DRLT), trade accounts payable (AP), and accrued expenses (XACC). We set the missing values of XRD and all missing changes to zero.

GPA - Gross Profits to Assets

Following Novy-Marx (2013), we measure gross profits to assets as total revenues (REVT) minus cost of goods sold (COGS) scaled by current total assets (AT), i.e.,

$$GPA_t = \frac{REVT_t - COGS_t}{AT_t}$$

O - Ohlson's O-score

Following Ohlson (1980), we measure O-score with the following linear relation

$$
\begin{aligned}
O_t = &- 1.32 - 0.407 \cdot \log(AT_t) + 6.03 \cdot \frac{DLC_t + DLTT_t}{AT_t} \\
&- 1.43 \cdot \frac{ACT_t - LCT_t}{AT_t} + 0.076 \cdot \frac{LCT_t}{AT_t} \\
&- 1.72 \cdot \mathbf{1}_{LT_t > AT_t} - 2.37 \cdot \frac{NI_t}{AT_t} - 1.83 \cdot \frac{PI_t + DP_t}{LT_t} \\
&+ 0.285 \cdot \mathbf{1}_{NI_t<0 \text{ and } NI_{t-12}<0} - 0.521 \cdot \frac{NI_t - NI_{t-12}}{|NI_t| + |NI_{t-12}|}
\end{aligned}
$$

where AT corresponds to total assets, DLC to short-term debt, DLTT to long-term debt, ACT to current assets, LCT to current liabilities, LT to total liabilities, PI to pretax income, DP to depreciation and amortization, and NI to net income. We winsorize all variables except for dummy variables at the first and 99th percentile of their respective distribution.

OPE - Operating Profits to Book Equity

Following Fama and French (2015), we measure operating profits to book equity as

$$
OPE_t = \frac{REVT_t - COGS_t - XSGA_t - XINT_t}{BE_t}
$$

where REVT corresponds to total revenues, COGS to cost of goods sold, XSGA to selling, general and administrative expenses, XINT to interest expenses, and BE to book equity. The definition of book equity is (a) the book equity of shareholders plus (b) deferred taxes and investment tax credit (Compustat item TXDITC or TXDB + ITCB if TXDITC is unavailable) minus (c) the book value of preferred stock. Regarding (a), the book equity of shareholders is shareholders' equity (SEQ), or the sum of common equity (CEQ) and the par value of preferred stock (PSTK), or, if all previous items are unavailable, total assets (AT) minus total liabilities (LT). Regarding (b), deferred taxes (TXDITC) or investment tax credits (TXDB plus ITCB) if deferred taxes is unavailable. Regarding (c), the book value of preferred stock corresponds in the following order either to the redemption value (PSTKRV), or the liquidation value (PSTKL), or if all previous items are unavailable to the par value (PSTK). We replace missing values of items (b) and (c) with zero. Moreover, missing values in COGS, XSGA, and XINT are set to zero.

ROA - Return on Assets

Following Balakrishnan *et al.* (2010), we measure return on assets as quarterly income before extraordinary items (IBQ) relative to lagged quarterly total assets (ATQ), i.e.,

$$
ROA_q = \frac{IBQ_q}{ATQ_{q-1}}
$$

Finally, the earnings announcement date has to be after the fiscal quarter end date.

ROE - Return on Equity

Following Hou *et al.* (2014), we measure return on equity as quarterly income before extraordinary items (IBQ) relative to lagged book equity (BEQ)

$$ROE_q = \frac{IBQ_q}{BEQ_{q-1}}$$

Quarterly book equity (BEQ) is computed as the (a) book equity of shareholders plus (b) balance sheet deferred taxes and investment tax credit minus (c) the book value of preferred stock. Regarding (a), we measure book equity of shareholders by shareholders' equity (SEQQ), or the sum of common equity (CEQQ) and the par value of preferred stock (PSTKQ), or if all previous items are unavailable by total assets (ATQ) minus total liabilities (LTQ). Regarding (b), the balance sheet deferred taxes and investment tax credit is TXDITCQ, or TXDBQ, or zero if both are missing. Regarding (c), the book value of preferred stocks corresponds to PSTKRQ, or PSTKQ, or zero if both are unavailable. Finally, the earnings announcement date has to be after the fiscal quarter end date.

TBI - Taxable Income to Book Income

Following Green, Hand, and Zhang (2013), we measure taxable income to book income as pretax income (PI) scaled by net income (NI)

$$TBI_t = \frac{PI_t}{NI_t}$$

We require positive pretax and net income.

IA-6. Size

ME - Size

Following Fama and French (1992), we measure size as the natural logarithm of the market equity. We obtain market equity data from CRSP by multiplying the shares outstanding (shrout * 1,000, as they are given in thousands) with the corresponding share price (the absolute value of altprc), i.e.,

$$ME_t = \log\left(shrout_t * 1000 * |altprc_t|\right)$$

IA-7. Trading frictions

AMI - Amihud Illiquidity Measure

Following Amihud (2002), we measure the illiquidity from daily return-to-volume (RV), which is the absolute daily return (r) scaled by the daily dollar trading volume (product of vol and prc), i.e.,

$$RV_t = \frac{|r_d|}{prc_t * vol_t}$$

Then, the Amihud illiquidity measure corresponds to the average return to volume estimate over the last six months. We require at least 50 observations for this average and adjust the trading volume of NASDAQ stocks according to Gao and Ritter (2010).

BETA - Beta Relative to the Market

Following Fama and MacBeth (1973), we measure the market beta β_1 from monthly CRSP data. Specifically, we run the following time series regression over the previous five years

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + u_t$$

Moreover, we require at least 24 monthly observations for the regression above.

DTV - Dollar Trading Volume

Following Brennan *et al.* (1998), we measure dollar trading volume from daily CRSP data as the average dollar trading volume (product of vol and prc) from month $t-6$ to month $t-1$ (with at least 50 observations). Moreover, we adjust dollar trading volume from NASDAQ according to Gao and Ritter (2010).

ISKEW - Idiosyncratic Skewness Relative to the Fama and French (1993) three-factor model

Following Bali *et al.* (2016), we regress daily excess returns in each month on the three Fama and French (1993) factor model (with a minimum of 15 daily observations), i.e.,

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + u_t$$

Then, idiosyncratic skewness is measured as the skewness of residuals u_t.

IVOL - Idiosyncratic Volatility Relative to the Fama and French (1993) model

Following Ang *et al.* (2006), we compute idiosyncratic volatility relative to the Fama and French (1993) three-factor model as the volatility of residuals from the following regression (with a minimum of 15 daily observations)

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + u_t$$

MDR - Maximum Daily Return

Following Bali *et al.* (2011), we measure the maximum daily (excess) return in each month from daily CRSP data (with at least 15 return observations).

TUR - Share Turnover

Following Datar *et al.* (1998), we measure the daily share turnover of each stock as the number of shares traded (vol) scaled by the number of shares outstanding (shrout), i.e.,

$$T_t = \frac{vol}{shrout}$$

Then, the share turnover for each firm in each month t is the average daily share turnover over the previous six months (with at least 50 observations). Moreover, we adjust the trading volume of NASDAQ stocks according to Gao and Ritter (2010).

IA-8. Valuation

BM - Book Equity to Market Equity

Following Davis, Fama, and French (2000), we measure the book-to-market ratio as book equity (BE) from Compustat divided by market equity (ME) from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$BM_t = \frac{BE_t}{ME_t}$$

The definition of book equity is (a) the book equity of shareholders plus (b) deferred taxes and investment tax credit (Compustat item TXDITC or TXDB + ITCB if TXDITC is unavailable) minus (c) the book value of preferred stock. Regarding (a), the book equity of shareholders is shareholders' equity (SEQ), or the sum of common equity (CEQ) and the par value of preferred stock (PSTK), or, if all previous items are unavailable, total assets (AT) minus total liabilities (LT). Regarding (b), deferred taxes (TXDITC) or investment tax credits (TXDB plus ITCB) if deferred taxes is unavailable. Regarding (c), the book value of preferred stock corresponds in the following order either to the redemption value (PSTKRV), or the liquidation value (PSTKL), or if all previous items are unavailable to the par value (PSTK). We replace missing values of items (b) and (c) with zero.

CFM - Cash Flow to Market Equity

Following Lakonishok *et al.* (1994), we measure the cash-flow-to-price ratio as income before extraordinary items (IB) plus depreciation (DP) scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$CFM_t = \frac{IB_t + DP_t}{ME_t}$$

We exclude all stocks with negative cash flows.

DM - Debt to Market Equity

Following Bhandari (1988), we measure the debt-to-market ratio as short-term debt (DLC) plus long-term debt (DLTT) from Compustat divided by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$DM_t = \frac{DLC_t + DLTT_t}{ME_t}$$

We exclude stocks with missing DLC and DLTT observations.

EBM - Enterprise Book Equity to Market Equity

Following Penman *et al.* (2007), we measure enterprise book equity scaled by market equity as net debt plus book equity scaled by net debt plus market equity

$$EBM_t = \frac{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t + BE_t}{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t + ME_t}$$

where DLTT corresponds to long-term debt, DLC to current debt, PSTK to the value of preferred stock, DVPA to preferred dividends in arrears, TSTKP to preferred treasury stock, CHE to cash and short-term investments, and ME to the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD). Lastly, in this case, book equity BE is computed as common equity (CEQ) plus TSTKP minus DVPA. Missing observations in DVPA and TSTKP are set to zero. Additionally, we require that the sum of net debt and book equity, as well as the sum of net debt plus market equity, are positive.

EM - Earnings to Market Equity

Following Basu (1983), we measure the earnings-to-price ratio as income before extraordinary items (IB) from Compustat divided by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$EM_t = \frac{IB_t}{ME_t}$$

We exclude firms with negative earnings.

NDM - Net Debt to Market Equity

Following Penman *et al.* (2007), we measure net debt to price as net debt from Compustat scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$NDM_t = \frac{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t}{ME_t}$$

where DLTT corresponds to long-term debt, DLC to current debt, PSTK to the value of preferred stock, DVPA to preferred dividends in arrears, TSTKP to preferred treasury stock, CHE to cash and short-term investments. Lastly, missing observations in DVPA and TSTKP are set to zero.

NPY - Net Payout Yield

Following Boudoukh *et al.* (2007), we measure the net payout yield as net payouts from Compustat scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$\begin{aligned} NPY_t = &[(DVC_t + PRSTKC_t + \Delta PSTKRV_t \cdot 1_{\Delta PSTKRV<0}) \\ &- (SSTK_t - \Delta PSTKRV_t \cdot 1_{\Delta PSTKRV>0})]/ME_t \end{aligned}$$

where DVC are dividends from common stock, PRSTKC is the purchase of common and preferred stock, PSTKRV is the value of the net number of preferred stocks outstanding, and SSTK reflects the sale of common and preferred stocks. $1_{\Delta PSTKRV<0}$ is a dummy variable that has value one if the annual change in PSTKRV is negative and zero otherwise. Moreover, we exclude stocks with negative net payouts.

OCM - Operating Cash Flow to Market Equity

Following Desai *et al.* (2004), we measure the ratio of operating cash flows to price as operating cash flows (OC) from Compustat divided by the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$OCM_t = \frac{OC_t}{ME_t}$$

Before 1988, we measure OC as funds from operations (FOPT) minus the change in working capital (item WCAP). Thereafter, we measure OC as net cash flows from operating activities (OANCF). Moreover, we exclude firms with negative operating cash flows.

SM - Sales to Market Equity

Following Barbee Jr *et al.* (1996), we measure the sales-to-price ratio (SM) as sales (SALE) from Compustat divided by the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$SM_t = \frac{SALE_t}{ME_t}$$

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